

# Jack Wei

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## EDUCATION

Olin College of Engineering, Needham MA - *Computer Science*

May 2028

Recipient of 4-year merit-based 50% Olin Tuition Scholarship

GPA: 3.96/4.0

Coursework: Data Structure & Algorithm, Software Systems, Computer Architecture, Machine Learning

## EXPERIENCE

Incoming Software Engineer Intern | Bloomberg

May 2026 - Aug 2026

Software Engineer Research Intern | Raytheon BBN Technologies

May 2025 - Aug 2025

- Refined **Python** data handler by integrating real-time processing of track quality, position error covariance, and confidence level via a customized message protocol
- Implemented system calibration using **MATLAB** to compute offsets on signal frequency, amplitude, & TDOA
- Presented software to **50+** stakeholders for a **\$1.5M** project funding opportunity

Mobile App Developer & Researcher | Olin PLai (AI) Lab

Jan 2025 - Present

- Designed & developed an iOS App in **Swift** that gamifies rehabilitation exercises for chair-bound or bed-bound elderly to exercise remotely using Computer Vision and AR glasses
- Achieved **95%** positive feedback rates at local Senior Center by integrating Multimodal AI Voice Agent using **LiveKit** to provide guidance on navigation of the interface and instructions on movements of the body
- Showcased product to **3,000+** attendees at Stanford Engineering Centennial

Software Development Research Intern | Raytheon BBN Technologies

Jun 2024 - Aug 2024

- Independently developed a large, distributed network management UI scalable for millions of government users using **Kotlin** and **Jetpack Compose**, and created a backend server using **Python Flask**
- Reduced **56%** of memory usage and **84%** of lookup time for network pairs by optimizing data architecture and streamlining backend processing of **OpenAPI**-defined API endpoints
- Spearheaded **5+** UI/UX design proposals & organized **3** team review meetings

UAS Synthetic Aperture Radar Researcher | MIT Lincoln Laboratory BWSI

Jul 2023 - Aug 2023

- Programmed and optimized radar back-projection algorithm, achieving a **12×** speedup using **Pythran**, **C++**, **NumPy Vectorization**, & **multi-threaded processing**
- Secured **96%** accuracy in object detections by implementing CFAR filters, median, and Gaussian smoothing
- Delivered a GUI supporting real-time signal data display and drone 3D path visualization using **wxPython**

## PROJECTS

Afora (Accountability for All) | <https://afora1.vercel.app>

Oct 2024 - Present

- Entered Babson Foundry Fellowship with **\$1,000+** funding and Butler Institute's Fast Track Program
- Implemented user-matching, rapid project outlining, and task distributing with a scoring system for teams using **Nextjs**, **Tailwind CSS**, & **Firebase**; integrated authentication using **ClerkJS**; deployed on **Vercel**
- Designed customizable UI with image search API and **Firebase Storage**; iterated on feedback from **100+** users

Factuality | <https://we1chj.github.io/FactualityPage>

Dec 2024 - Jan 2025

- Won **1st** place in the Generator Entrepreneurship & AI Buildathon with **\$2,500+** project funding
- Developed & published a Chrome extension in **React** & **Tailwind CSS**, featuring easy fact-checking by integrating AI web-scraping for citations and sources by prompt-engineering via **ChatGPT** & **Perplexity** API

## HONORS

International Collegiate Programming Contest – Top 5 in District

2025

USA Computing Olympiad Gold Division – Top 500 Pre-College Students Nationally

2023

American Computer Science League – Two-time National Finalist

2023

Google Kickstart 2022 Round C – Ranked Top 2% Globally

2022

## SKILLS

**Frontend:** JavaScript, TypeScript, Nextjs, Tailwind CSS, Kotlin Multiplatform, Jetpack Compose, ThreeJS

**Backend:** C/C++, Python, Java, Kotlin, Flask, Firebase, Supabase, Spring Boot, SQL, MongoDB

**Deployment:** Git, Vercel, Bitbucket, Apache Maven, Gradle

**Misc:** MATLAB, Swift, Bash, OpenAPI, LiveKit, Unity, C#, Bash, Blender, Vim, Docker, ROS2